Worksheet (Sets)

Course: Discrete Mathematics

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1.

Use a membership table to show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

2.

Use set builder notation and logical equivalences to establish the first De Morgan law $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

3.

Determine whether each of these sets is the power set of a set, where a and b are distinct elements.

a) $\emptyset$
b) $\{\emptyset, \{a\}\}$
c) $\{\emptyset, \{a\}, \{\emptyset, a\}\}$
d) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

4.

How many elements does each of these sets have where a and b are distinct elements?

a) $\mathcal{P}(\{a, b, \{a, b\}\})$
b) $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$
c) $\mathcal{P}(\mathcal{P}(\emptyset))$

5.

What is the cardinality of each of these sets?

a) $\emptyset$
b) $\{\emptyset\}$
c) $\{\emptyset, \{\emptyset\}\}$
d) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$

6.

Let A, B, and C be sets. Show that

$$\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}.$$

7.

Let $A_i = \{1, 2, 3, \ldots, i\}$ for $i = 1, 2, 3, \ldots$. Find

a) $\bigcup_{i=1}^{n} A_i$.

b) $\bigcap_{i=1}^{n} A_i$.

8.

Let $A_i = \{\ldots, -2, -1, 0, 1, \ldots, i\}$. Find

a) $\bigcup_{i=1}^{n} A_i$.

b) $\bigcap_{i=1}^{n} A_i$.

9.

Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of these sets with bit strings where the ith bit in the string is 1 if i is in the set and 0 otherwise.

a) $\{3, 4, 5\}$

b) $\{1, 3, 6, 10\}$

c) $\{2, 3, 4, 7, 8, 9\}$

10. The successor of the set A is the set A ∪ {A}.

Find the successors of the following sets.

a) $\{1, 2, 3\}$

b) $\emptyset$

c) $\{\emptyset\}$

d) $\{\emptyset, \{\emptyset\}\}$